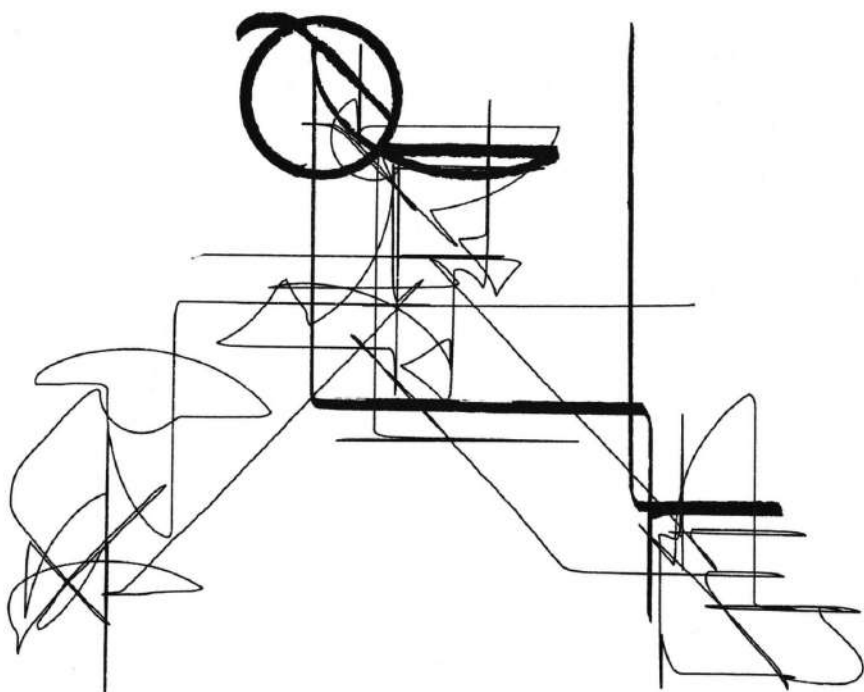


Issue 1: Human Computer Interaction



Machine
zine

Letter from the Editor(s)

Welcome to the inaugural edition of Machine Zine!

To make a zine about human computer interaction in 2026 is, almost inevitably, to make a zine in the shadow of generative AI.

The current cultural language around AI often encourages mystification. Systems are described as if they think, imagine or create with autonomy. Rather than accepting the machine as an opaque source of wonder, these contributors examine it as a material, social and cultural system.

This zine is about machines as systems of constraint, negotiation and perception. It is not just about AI, but the longer cultural and artistic history of machines: analogue computers, drawing machines, ZX Spectrum sound, web identity, image generation, sensors, rooms, rules, interfaces and artistic practice.

Machines have never simply helped people make things. The machine as frictionless assistant has never existed. In practice, machines shape the conditions of making imposing systems that inevitably mark the work produced through them. This is true whether the work takes the form of an image, a sound, a drawing, a performance, an archive, an avatar or an online persona.

The zine is about human-machine relations after the fantasy of frictionless technology. It argues across articles that machines are not neutral tools. They are systems with histories, biases, material behaviours and cultural effects.

Machines are not external to culture. They are cultural systems. Artists should not merely use them, but test them, resist them, expose them and work with their limits.

-Sean Carroll

Table of Contents

EDITOR'S LETTER - 3

CONTRIBUTORS - 5

THE MOST INTERESTING MACHINES SAY 'NO', ANDY JEN-
KINSON - 6

ARCHITECTURE OF SUSPENDED NETWORKS AND IDENTITY

BOUND, DENISE PARDON BENITEZ & SEBASTIEN DURAND - 9

WE MUST ALL BE ARTISTS NOW, SEAN CLARK - 13

FRUITS OF LABOUR, SAVANNAH IRVING - 14

AN ARTISTIC COLLABORATION BETWEEN MAN AND MACHINE,

PAUL MAZZITELLI - 17

WE STAND BEFORE THE MACHINE: ROOMS, RULES, SYSTEMS,

AND THE THINKING MACHINES THAT ARRANGE US, TIMOTHY

ELLIM (TIMOTHY LIM) - 21

Contributors

ANDY JENKINSON

[HTTPS://BUTTERFLYPOLITE.COM/ANDY/INDEX.HTML](https://butterflypolite.com/andy/index.html) ANDY JENKINSON IS A LEICESTER-BASED SOUND/MULTIDISCIPLINARY ARTIST WHOSE PRACTICE EXPLORES LEGACY TECHNOLOGY AND CONSTRAINT, INVESTIGATING HOW CREATIVITY EMERGES WHEN ARTISTS PUSH MACHINES BEYOND THEIR INTENDED LIMITS.

DENISE PARDON BENITEZ & SEBASTIEN DURAND

SINCE SEPTEMBER 2023, ARTIST-DUO SÉB AND DENISE HAVE EXPLORED VARIOUS ASPECTS OF THE HUMAN PSYCHE, DIGITAL MEDIA AND EXPERIMENTAL STORYTELLING IN THEIR PROJECT EXERCITIA SPIRITUALIA ANIMAE. [HTTPS://JOURNAUXINDIENS.NET/](https://journauxindiens.net/) [HTTP://PADRON-BENITEZ.NET/](http://padron-benitez.net/) [HTTPS://EXALTATIO-ANIMAE.NET/](https://exaltatio-animae.net/)

SEAN CLARK

SEAN CLARK IS AN INDEPENDENT ARTIST, CURATOR, AND RE-SEARCHER BASED IN LEICESTERSHIRE, UK. HIS ARTWORK EXPLORES INTERACTION AND CONNECTEDNESS THROUGH THE CONSTRUCTION OF AUDIOVISUAL SYSTEMS PRESENTED ON SCREEN, AS INSTALLATIONS, AND AS PRINTS.

SAVANNAH IRVING

MY ARTISTIC PRACTICE TAKES A MULTI DISCIPLINARY APPROACH TO MAKING WORK, EXPERIMENTING WITHIN DIGITAL AND ANALOGUE MEDIUMS LAYERING VISCERAL TEXTURE AND SATURATED COLOUR TO DISTORT AND RENDER SUBJECT MATTER.

PAUL MAZZITELLI

MY PRACTICE IS CYBERNETIC, INTERDISCIPLINARY AND COMPUTATIONAL. IT IS CYBERNETIC IN ITS ENGAGEMENT WITH SYSTEMS ART, FEEDBACK, SYSTEMS THEORY AND BEHAVIOUR. I AM INTERESTED IN HOW STRUCTURES REGULATE THEMSELVES, HOW INFORMATION CIRCULATES, AND HOW CONTROL AND ADAPTATION EMERGE THROUGH INTERACTION. I AM CURRENTLY STUDYING FOR AN MA IN FINE ART AT DE MONTFORT UNIVERSITY.

TIMOTHY ELLIM (TIMOTHY LIM)

TIMOTHY LIM IS A PRACTICE-BASED PHD RESEARCHER WHO WORKS ACROSS GAME STUDIES, PERFORMANCE, AND CREATIVE TECHNOLOGY. HIS PRACTICE EXAMINES HOW INTERACTIVE SYSTEMS SHAPE COOPERATION, MEANING-MAKING, AND PUBLIC EXPERIENCE IN CO-LOCATED ENVIRONMENTS.

The Most Interesting Machines Say ‘No’

ANDY JENKINSON

THE PROMISE OF FRICTION-LESS TOOLS

Creative technology now measures its success by how little resistance it offers. Intuitive user interfaces, optimised human-computer interaction, and seamless workflows promise less time spent mastering tools and more time spent on the work that matters: writing the book, making the music, drawing the picture.

Adobe, Apple, Unity, and the rest sell ease-of-use as a virtue. Presets and pre-built assets let you begin immediately; producing something recognisable within minutes. Basic tools handle everyday tasks, while increasingly complex features cater to the ‘power user’. Many now even offer AI suggestions for what you might want to do and how best to do it.

These experiences are often framed in terms of “cognitive ergonomics.” Reduce mental load. Remove obstacles. Enter a state of flow where the tool disappears entirely and the work pours out, effortlessly reflecting your intent.

Bollocks.

Friction is not a flaw. Friction is not a problem. To paraphrase Marshall McLuhan, friction is the medium.

Art emerges at the point we push against the constraints of the medium - or where it pushes back against us.

MACHINES THAT SAY NO

I contend with this in my artistic practice when writing new compositions for the 1982 ZX Spectrum, where every musical note exists only because a loop of instructions flips a tiny speaker on and off thousands of times a second. This machine offers no shortcuts - no sound chip, no timing circuitry - the pitch of notes is determined by the length of time the actual computing instructions take to operate.

Often I find myself attempting to achieve something within this system that breaks something else.

Adding a routine to check for keypresses might change the tim-

ing or pitch of a note. It becomes an exploration of what is possible versus what I can ‘get away’ with.

As I wrestle with this, I notice subtleties in the system’s flaws, such as the poor frequency response of the ZX Spectrum’s speaker. I tweak my music to exploit these, hiding complexities in the places where the limitations of the computer can ‘do less damage’. The clock speed of the computer and the efficiency of my code limit the range of notes I can produce. Internal registers being only 8-bit compromise the accuracy of tuning loops on high-pitched notes. Solutions present themselves: I move to 16-bit register-pairs, but the extra computation time limits the maximum pitch. These micro-frictions shape the choices I make in the composition itself.

My frustration at the computer’s lack of compliance is palpable.

My creative process is not a flow-state of music emerging unhindered from my soul - it is about overcoming barriers, negotiating, and making compromises over and over again.

The computer does its best to sabotage my artistic vision, and I adapt my work in response.

WORKING AGAINST THE MACHINE

My practice often involves working with technologies that are awkward, limited, and often deeply unhelpful: the aforementioned ZX Spectrum computer, Teletext graphics systems, and GameBoy music software.

These machines do not behave like modern creative tools. They refuse convenience. They impose strict limits and strange rules. They require workarounds, hacks, and invention.

Creating Teletext art isn’t simply a matter of placing pixels on its 80x72 grid, as it may outwardly appear. Foreground and background colours can’t be changed simultaneously, and each change imposes a compromise: a 2x3 pixel gap of background colour, or a forced repetition of the previous 2x3 pixels to the left. The technical barrier to make a piece is low - placing the pixels is easy.

But fighting the restrictions is where the art emerges.

The constraints shape creations: refusing left-right symmetry, suggesting new framings, perspectives, and approaches to a piece. The Teletext art ‘scene’ may be small, but I can recognise other practitioners’ work through the way they have navigated the systemic limitations - the stylistic

scars left from their battles with the arcane conventions of the medium.

NEGOTIATION: NOT CONTROL

Human-computer interaction is normally framed as a problem of usability. Systems are made faster, smoother, and more intuitive. That assumes ease is the goal.

I actively choose to work with systems where usability is not at the forefront – not out of nostalgia or masochism – but because my interactions become something different entirely. They become a collaboration.

Collaboration isn't obedience. Resistance is an integral part of any bilateral process; Artist vs. Machine, with both pushing each other to improve. It is a combination of stubbornness, alternative approaches, and repeated failure, with both contributing ideas that inform the finished work.

THE PROBLEM WITH HELPFUL MACHINES

Given no constraints, art atrophies.

The option of every paint colour, every instrument, and an infinite supply of Lego bricks leads to paralysis. Work is never completed. Art could always be “improved.”

“The only limits are your imagination.” This is a preposterous lie.

Art should challenge. And artists should be challenged.

Every upgrade to a piece of creative software gives new tools and new conveniences; a new toy that an artist can fight with, and possibly create something ‘negotiated’ within that first flush of exploration. But it soon becomes a trivial way of achieving something that might previously have been an arduous process – an arduous process that added to the art. Because of the machine’s artefacts.

SEEK DIFFICULT MACHINES

It is not enough to reject the latest update after having sampled its slicker workflow. This poisons the previously beautiful collaboration with the uncooperative machine. It reframes it as a failing, and the user starts resenting the process as a means to a goal. This is not art.

It is not enough to simply not explore the tools available to you. This is to wallow in ignorance, and it becomes a pastiche of naivety. This is not art.

It is imperative that the Artist and the Machine make demands of one another.

This is only possible by rejecting ease.

Choose difficult machines, systems, and tools. Embrace their limitations and constraints as part of the artistic process, and as your collaborator in the work.

Choose machines that say ‘no’.

Architecture of Suspended Networks and Identity Bound

DENISE PARDON BENITEZ &
SEBASTIEN DURAND

Signals fizz beneath the dermal network
 layers.

What was once imagined as a space of radical openness now hums in dispersed, islanded formations. The transition from the early web to Web 2.0 marked an ontological shift. The open, text-based web gave way to enclosed, profile-centric environments in which identity is modular, persistent, and continuously negotiated. In Moeller and D'Ambrosio's *You and Your Profile* is “profilicity” described as a condition in which the notion of a coherent, authentic core self dissolves.¹ Rather than reflecting identity like a mirror, the profile refracts it – a prism scattering the self into multiple projections, recalling ancient figure of Proteus, the ever-shifting sea god who resists to be captured. In the social age, the Protean condition of mutability became infrastructural as network capacities grew and “connected people”.

To grasp him was to lose him.

The profiled identity enables to curate the self in alignment with platform logics. It operates under conditions of second-order observation: the subject observes itself being seen. Interactions between profiles produce feedback loops of validation, where engagement functions as both measure and incentive. The Sophists argued that truth is not essence but appearance and persuasion. What matters is not what one is, but how one appears to others. The shift from being to being-seen: identity as an operation performed under the gaze, constructed in anticipation of interpretation.

Within the current shift from web2.0 to the hyperconnected web3.0 landscape emerges *PHERSU* (*phersu.net*), a net-art project engaging HTML5 as both medium and method. Through identity display, fragmented environments, and algorithmic processes, it explores the instability of digital subjectivity. As the second chapter of *Exercicia Spiritualia Anima*,² the socially constructed identity is enacted in a world where it cannot be stabilised but deformed and misaligned by the very systems meant to contain it. Labyrinth-like and open-ended.

Although the fluidity of identity is governed online by its very antithetical practice: “You need to verify your profile” – click a link, copy-paste a series of numbers – Identity becomes a matter of consistency, repetition and linking. Verification systems demand coherence across data points. As artist and activist Heath Bunting demonstrates in his work, the legal self is assembled through linked information—forms filled, records aligned—rather than lived continuity. If enough data-points appear over and over again, you are proven to exist.



PHERSU renders this logic visible. *ID Cards* is a page that presents a stenographic reading of administrative language, an ode to bureaucratic structure that borders on absurdity. Its seriousness exposes itself as tonal—capable of slipping into something like ambient noise, a background hum of governance. Registration and categorisation of people is as old as civilisations and star dust, and has its history of domination and vulnerability. These days, the self-involved mindless entertainment to present oneself for an oracle-like online test also involves leaving a trace of one's soft inner core for the mongers to feast on, may it be purely absurd/ha-ha funny or may it be medical self-diagnosing. The seeking for a spoon of self-indulgent soup served by the machine is the theme of *Personality test*⁴, a HTML as a generative source of poetic touch, reflecting on the desire to know and the unpredictable outcome.

The condition of prolificity also intensifies the fragmentation of the self described by Fredric Jameson.⁵ The subject no longer occupies a singular role but is split and distributed across platforms: banking, employment, social media. Each requires a distinct performance. Life becomes a sequence of accounts, each secured, separated, and maintained. The unified self dissolves under this multiplicity, revealing itself as a retroactive construction.

In PHERSU, this mutability extends to navigation and perception. *Reflected Immediacy*⁶ renders a video through itself into abstraction, oscillating between interior and exterior states—a visual echo of Hegel's inner/outer dialectic, where both are moments of the same present. *Kitchen*⁷ presents a suspended 3D scan accompanied by looping affirmations (warped), staging the contemporary ritual of self-optimisation: to think oneself into success, love, coherence.

The hypertext, “will typically be non-linear, branching, and large with various options to the user” wrote Theodor H. Nelson 1965, and adds further that this new media text then, will make way for the natural curiosity and self education, bringing together and crossing many sources with each other.⁸ *Wikipedia*⁹ is a page built on a simple idea of brining Laconia-related articles into a network map allowing to dig logs and history produced by anonymous historians and individual editors (propon) around the world, revealing connections that one might find educative, surprising or

sad. An experimental mapping that both could find its way through journalism and research, either to illustrate geo-political disturbance as well as giving all the characteristics of the peaceful land as antique warriors that experienced exile in the region.

Wiki-sites are, in a way, a vestige of by-gone era of the web. The practice of co-authored text-based pages, where hive-minds built on each other's contributions were occurrences in the early social web. The root of these practices are described in *Mapping Cyberspace*; how the early text-based MUDs [Multi User Domains] were a real-time collectively written novel that unfolded over time.¹⁰ To be inside the text instead of an outside reader: a virtual role-play where each user could participate into the world-making, a process which involved both coding and fiction writing. The appearance of these social spaces changed and became virtual spaces as network capacities and computer graphics grew more efficient towards the end of the millennia. For the contemporary reader, the social and creative practices can be encountered in Sherry Turkle's *Life on the Screen*¹¹ or the the 2003 documentary *Online Traveller: AVATAR*¹², in which room-creation is done both with the collective in mind as well as to express or externalize parts of one's self. An memory of the web before the platform demanded you to appear with your legal name, when you could be anyone behind and alias and, in some cases, an unbound body of an avatar, which term itself carries a longer history.

Proto-Indo-European au- (“ o f f , a w a y , d o w n ”) a n d terh₂ - (“to cross over, pass through, overcome”) → Sanskrit ava (“down, off”) + tarati (“crosses over”)

1784– Hinduism: manifestation of a god in bodily form,

1815– embodiment of a principle or force,

1986– computing: graphical representation of a user

From divine descent to digital proxy, the avatar marks a shift in how presence is constructed. Online, this elasticity enables experiments with gender, anonymity, and relationality, forms of i n t i m a t e a n o n y m i t y a n d a n o n y m o u s i n t i m a c y. At the same time, distinctions blur in the opposite direction. Bots are trained to appear human; humans perform robotic patterns. The boundary between subject and system destabilizes. It is not implausible that individuals will come to operate as corporations, while corporations simulate trusted persons. At the same time, distinctions blur in the opposite direction. Bots are trained to appear human; humans perform robotic patterns. The boundary between subject and system destabilizes. It is not implausible that individuals will come to operate as corporations, while corporations simulate trusted persons. .You can be anything this time around.

Yet platform templates produce serialized identities: users respond to standardized prompts, align with circulating aesthetics, reproducing recognizable formats. Customisation becomes endless project within constraint, such as prefabricated architecture; variation exists, but structure remains fixed. Within this suspended architecture, identity is neither fixed nor free.

It is orchestrated through constraints, shaped by invisible thresholds, and contingent on systems of recognition. The web 2.0 doesn't want their users to get lost in world-making, instead there is a demand to exist on a platform in the exact same way as everyone else. What persists is not the self,

but its trace-indexed, filtered, and conditionally allowed to remain.

Against this, recent writings such as “Welcome to the Post-Naive Internet Era”, there is a call to

explore how to rebuild online infrastructures grounded in communal values rather than extractive systems.¹³ Yet naivety can be reclaimed—not as ignorance, but as exploratory force. If the “internet native” operates fluently within systems, the “internet naive” probes their structures, moving without prescribed limits of expression.

The fragmentation does not only produce loss. It also opens another trajectory: not toward unity, but toward a different coherence. In certain strands of ancient thought, the self is not singular but an aggregation - a flux without centre. To recognize this is not collapse, but release of the Tecnoetic.¹⁴

What emerges is the possibility of a distributed awareness: a subject capable of inhabiting multiplicity without being reduced to it. If Web 1.0 enabled exploratory presence, and Web 2.0 produces the serial self,

then beyond this lies a potential reconfiguration: a networked consciousness that holds contradiction without resolving it. Between code and cognition, between fracture and flow, something persists. Not an escape from systems, but a navigation within them. A protean condition, no longer imposed but enacted.

Footnotes

- 1 Moeller, Hans-Georg & D'Ambrosio, Paul, *You and Your Profile Identity after Authenticity*, 2021. 2 Exercicia *Spiritualia Animae* is a tritych project exploring the human psyche through ancient mythology, alechemy, expanded media and technology. <https://exaltatio-animae.net/>
- 3 https://phersu.net/id_card
- 4 https://phersu.net/personality_test
- 5 Jameson, Fredric, *Postmodernism, or, the Cultural Logic of Late Capitalism*, 1991.
- 6 https://phersu.net/reflected_immediacy
- 7 <https://phersu.net/kitchen>
- 8 Nelson, Theodor H, “The Hypertext” <<https://archive.org/details/thehypertext/TheHypertext/>>
- 9 <https://phersu.net/wikipedia>
- 10 Dodge, Martin & Kitchin, Rob, *Mapping Cyberspace*, 2001, p. 146
- 11 Turkle, Sherry *Life on the Screen*, 1995.
- 12 Mancini, Donato & Turner, Jeremy, *Online Traveller: AVATARA* <https://archive.org/details/vw_traveler-avatarax> 2003.
- 13 Matusek, Severin, Houde, Nic and Moniz, Paloma “Welcome to the Post-Naive Era” <<https://www.mozillafoundation.org/en/nothing-personal/the-post-naive-internet-era/>> [23-04-2026]
- 14 Ascott, Roy, *Art, Technology, Consciousness*, 1996.

We must all be Artists now

SEAN CLARK

To thrive in a world increasingly shaped by AI and robotics, **we all need to think more like artists.** This does not mean that everyone must become a professional practitioner in the arts, but that we must cultivate an artistic way of engaging with the world: one grounded in curiosity, experimentation, reflection, and a willingness to question assumptions.

It also requires rejecting the idea that the technologies we use are neutral, inevitable, or fixed. Every tool carries values, biases, and intentions, and part of thinking like an artist is learning to recognise, interrogate, and reconfigure those forces rather than passively accepting them.

We must learn to value play as a serious mode of inquiry, not as a distraction.

Play enables us to test possibilities, invent problems, and explore alternative futures without needing immediate justification or utility. Alongside this, we need processes of making that are structured and knowledge-led, yet open enough to accommodate chance, interpretation, and imagination. It is within this tension between structure and openness that genuinely new ideas emerge.

As humans, we possess lived experience, embodied perception, memory, culture, and situated knowledge that cannot be replicated or truly inhabited by machines.

Even as AI systems grow more sophisticated, they do not experience the world; they process representations of it.

Our unique perspectives allow us to interpret, contextualise, and reflect in ways that are inseparable from our histories and our physical and social existence. No matter how persuasive claims about machine understanding may become, this fundamental distinction remains.

If we are to take meaningful control of our future, we must focus on what differentiates us from AI rather than competing with it on its own terms. Our capacities for critical thinking, ethical judgement, empathy, and self-reflection are not secondary qualities; they are central. So too are authenticity and integrity - the ability to stand behind our choices, to articulate why we make what

we make, and to accept responsibility for the consequences.

Thinking like an artist, then, is not a luxury for a cultural elite. It is becoming a necessary survival skill.

We can use technology as a material for expression, inquiry, and **amplification** of human values by embracing artistic modes of thinking, rather than allowing it to become a substitute for them. In doing so, we assert that the future is not something that simply happens to us, but something we actively shape.

Fruits of Labour

SAVANNAH IRVING

'The evolution of human intelligence is primarily based on information we have gathered through vision, light has come to mean illumination and enlightenment.'

- Lynn M. Herbert

The point at which each of my subjects transforms within the digital reveals a complex duality, mirroring biological patterns of rebirth and disintegration. Through investigating the tension between life and decay, the sacred and synthetic each iteration of work leads to the production of forms that resist singularity, presenting an intimacy that feels almost alien. My practice seeks to explore instabilities found in the virtualisation of contemporary image culture. Working with digital distortion to question how bodies, landscapes and ecologies are constructed and mediated.

Inspired by 3d scans of religious symbolism taken during my time in Spain. My most recent project, Nature Morte, investigates the awe-inspiring intricacies of natural forms, Baroque sculpture and Romantic philosophy within the context of emerging media.

These references are translated into a visual language where photo manipulation subverts traditional convention, reinterpreting sensations associated with the natural world within technological systems of image creation. As

this process has developed, the complexities that lie behind this digital sublimity become more intriguing.

The sense of limitlessness that exists at the surface of each of our screens presents an almost delightful horror. Light has come to not only exist as a beacon of vision, but in an age of information becomes a gateway towards transcending the material. When engaging with this hyper-objectivity we are met with the inadequacies of our own perception. This anomaly is particularly evident when observing our accelerated consumption of AI generated imagery. As I continue to move back and forth between organic material and digitally rendered forms, the paradoxical nature of my own work begins to mirror this optical occlusion. Within this process, light becomes more than a tool but a material force that unveils texture, depth and impermanence.

This relationship between the real and the unreal as well as what occurs at the intersection of this symbiosis, replaces photorealism with illogical detail. The hallucinatory aspect of



*Fantasy
In too deep
Made to last, made to crash
Show me what you're...*

this imagery creates an almost tangible world condensed into the confines of the familiar rectangular format, which has become the dominant object of our awareness. Within this mediated cognitive experience a vague yet all-encompassing question arises: What is freedom? Is it the ability to indulge in all of the senses or a lack of? This nuanced relationship between human and machine has developed beyond utopian narratives of unification. Within my own practice, this negotiation between autonomy and control is not resolved but rather reflected on.

As an artist exploring the nuances of digitally constructed imagery, perception becomes medium. Refraction, emission and specular subsurface, the uncanny nature of each final image delivers illusory forms that feel slightly discernible. This dissonance converses with a broader discourse around artificial representation, where reality is no longer defined by what we know to be authentic but by the capacity to simulate, respond and be perceived as living. In the case of generative AI, this nuance between the monetary and emotional value of embodiment actively affects natural environments exemplifying this conflict between two worlds.

Reframing my own connection to the natural world as an interdependent relationship where the restraints of my environment become a wellspring of inspiration to pull from, presents an urge to engage with a source of infinite discovery. The challenge of re-representing these forms through a formulaic medium reflects a desire to balance these two opposing forces, co-creating a sense of beauty that occupies space within the unsettling aspects of the uncanny.

As this archive of imagery morphs further into abstraction, the transient nature of this referential way of working presents a time-based process of evolution, mimicking the perpetual unfolding of natural phenomena, ultimately leading to a collapse in figuration. Pushing the boundaries of this aesthetic experience disrupts a stable sense of the real.

Within virtual bio-mimetics, these environments produce forms that shift, respond and resist categorisation. As this mode of working progresses, my intention as an artist is to represent the labour that exists behind each image, illuminating the cracks, blemishes and errors that occur during this push and pull between preservation and erosion. Choosing to work in this way allows for an embrace of limitation, through constraint these outcomes become unpredictable approaching each piece as an artefact of experimentation.

The glitch within this workflow subverts the high gloss tactile aesthetic of these forms, a process which actively invites viewers to question their own recognition. Capturing the faults of digital memory not only highlights the value found in imperfection, but allows for further conversations around aesthetic representation. Re-examining how themes such as intimacy, identity and ecological connection manifest within a coded space, reinterprets digital boundaries as a place for artistic opportunity.

An artistic collaboration between man and machine

PAUL MAZZITELLI

At its most basic level, digital computation is not abstract intelligence. It is sequencing/timing (CALCULATED EVENTS). Beneath graphical interfaces and complex learning systems lies a continuous stream of electrical signals switching on and off at extraordinary speed. Binary code is simply the organisation of these timed pulses, sequences of voltage changes structured into patterns. Computation is the coordination of rhythm.

If this is the case, then the distinction between digital and analogue systems becomes less categorical. A mechanical device that regulates phase, delay and repetition is also organising time. Rotational speed, gear ratios and resistance form patterned sequences, just as clock cycles and logic gates do. The difference is one of scale and material, not principle.

What, then, qualifies as a computer?

Is it defined by silicon, or by the structuring of timed relationships? The word “computer” originally referred to a person, often a woman, employed to calculate tables and process numerical data. Computation was once a form of labour. The machine inherited that role, embedding timing and sequencing into hardware.

Jack Tait’s programmable analogue drawing machines, known as Taitographs, can be understood within this expanded definition. Developed from the 1960s onward using repurposed electro-mechanical components, these systems generate intricate drawings through differential gearing, phase relationships and calibrated instability. Their logic is not written in code but expressed through motion. They compute through voltage, mechanical restraint and chance.

Through sustained engagement with these machines at the Computer Arts Archive and R10.EMS, I have come to

recognise interaction as participation within structured constraint. Tait's drawing devices operate through calibrated timing and subtle irregularities, nuances and quirks that give each machine its character, much like the sequencers, drum machines and synthesisers that shape my electronic music practice. In each case, authorship emerges not from total control but from improvisation within a bounded system, an artistic dialogue between human and machine.

Jack Tait's work emerges from a distinctly British lineage of artist-engineers who explored computation through material systems during the mid-twentieth century. Among his early influences was the philosopher and drawing-machine pioneer D. P. Henry, whose mechanically programmed works of the 1960s demonstrated that generative form could be achieved through engineered motion. Henry's approach of adapting a bomb-sight computer and precision mechanical components into autonomous drawing systems revealed the aesthetic potential of analogue calculation. Tait encountered this work at a formative moment in his own development.

A key catalyst was his visit to Cybernetic Serendipity in 1968. The exhibition, curated by Jasia Reichardt, brought together early experiments in computer art, robotics and systems-based practice. It presented computation not simply as industrial technology, but as a creative medium. For Tait, the exhibition clarified a possibility: that his early experiments with Meccano and motors were, in fact, in line

with, and perhaps even stronger than, other machines on show at Cybernetic Serendipity. From the late 1960s onward, Tait began constructing his own programmable analogue drawing machines using repurposed electro-mechanical components. Unlike digital computer artists working with mainframes, Tait embedded programming directly into gears, cams and rotating assemblies. Rather than simulating mathematical form, his machines enacted it physically, producing images as the residue of motion.

Over several decades, Tait refined and expanded these systems, developing increasingly sophisticated mechanisms capable of complex graphic outputs. His work has since been recognised as an important contribution to early generative practice. Taitographs and associated machines are now held in institutional collections, including the Science Museum, situating his analogue experiments within a broader history of computational art in Britain.

Tait's drawing machines are constructed from repurposed electro-mechanical components. Each operates through differential gearing and phase offset, where multiple rotational elements interact to produce complex, evolving patterns. Small adjustments to speed, resistance or alignment can significantly alter the resulting image. The drawing is not predesigned but emerges from the interaction of timed mechanical relationships.

Several of the systems are structured around what Tait refers to as "Sun" and "Moon" mechanisms,

paired rotating elements whose relative motion produces shifting orbital patterns. As one disc rotates against another at differing speeds, their phase relationships continually evolve, generating intricate geometries that expand, collapse and reconfigure over time. These celestial metaphors are not decorative; they describe the logic of the system. The drawing is the trace of rotational negotiation.

Crucially, Tait has often stated that the machines themselves are not the artwork. For him, the artwork resides in the Taitograph, the drawing produced through the machine's output. The device is a means of structuring motion, a programmable apparatus through which form is generated. The emphasis remains on the drawn image rather than the object that produces it.

Yet it is difficult to separate the image entirely from its mechanism. The machines possess a sculptural presence: exposed gearing, rotating assemblies and calibrated tension systems, sometimes partially housed in opaque perspex, that render their operation visible. They reveal computation in motion. Many viewers, myself included, regard them not simply as tools but as mechanical sculptures in their own right, systems that both generate and embody aesthetic logic. The drawing may be the final artefact, but the machine is the performative body.

Central to their operation are the controllers. These function as physical interfaces, enabling negotiation with the system through variable speeds, couplings and mechanical con-

figuration. Interaction begins with setup choices, pen type, paper, controller alignment, but prescribed and improvised adjustments during the drawing process can result in dramatically different visual outcomes. The so-called "chance" controllers introduce further instability, allowing subtle mechanical variations to shape the final composition.

In later years, Tait incorporated digital processes into his workflow, scanning drawings and refining them using software such as Photoshop. This did not replace the analogue core of the work but extended it, allowing further exploration of colour, layering and tonal variation. The digital stage becomes a secondary transformation of mechanically generated form.

The machines therefore operate across a threshold: analogue in generation, occasionally digital in post-production, always structured by mechanical timing. Whether regarded as instruments, sculptures or computational artefacts, they remain deeply engaging to work with. A mechanical musical performance is a byproduct, the sound of motors and rotating assemblies marking time as the image gradually emerges.

During our conversations about the future of the machines, Jack posed a question that sits at the centre of this project: if another artist were to use the same systems, operating within the same mechanical constraints, would the resulting drawings be meaningfully different? The answer, in practice, is yes.

Although the machines operate through fixed structural logics, the space of

possible variation is far from exhausted. Decisions about configuration, speed relationships, pen selection, paper type and duration all shape the outcome. Subtle interventions during operation, when to introduce resistance, when to allow drift, when to stop, alter the evolving pattern. Within constraint, interpretation emerges.

My own drawings produced on Tait's systems differ markedly from his. The mechanical vocabulary remains the same, yet the pacing, density and tolerance for instability shift. Where one configuration might prioritise symmetry and closure, another may extend duration and allow irregularity to accumulate. The machine provides a structure; the artist determines how long to remain within a given creative relationship and when to intervene.

A useful parallel can be found in electronic music production. For over four decades, the Roland TR-909 drum machine has formed the rhythmic foundation of countless techno records. Its architecture is fixed: the same kick drum, the same hi-hats, the same sequencer grid. Yet a seasoned listener can distinguish immediately between a track by Jeff Mills, The Mover or Neil Landstrumm by the rhythmic foundations of their productions. Despite the shared instrument, the machine does not erase authorship; it frames it.

Tait's drawing machines function similarly. They establish a bounded system, a mechanical grammar. Within that grammar, interpretation occurs through timing, emphasis and restraint. Authorship does not reside solely in the invention of the apparatus, nor solely in the output, but in the negotiation between structure and decision.

The question of whether the machine determines the drawing dissolves when viewed this way. The system constrains, but it does not prescribe. Just as the TR-909 does not dictate the aesthetic of a track, the Sun and Moon mechanisms do not dictate the aesthetic of a Taitograph. They offer a field of possibility, and it is the role of the artist to explore these interrelationships between structure and creativity.

After years of working with synthesisers, sequencers and drum machines, I have come to recognise the same structural logic in Tait's drawing machines. Each is a system that unfolds in time. Each resists total control. Each reveals its character through constraint.

The Taitograph on the cover of this issue is not simply an image. It is the trace of regulated motion, a collaboration between rotating elements and human decision. It records a period of negotiation within a mechanical system. In that sense, it is no different from a recorded performance on a drum machine or modular synthesiser. It is a document of timing. If we widen our understanding of what a computer is, and what interaction entails, then Tait's machines sit comfortably within that history. They demonstrate that computation can be mechanical, visible and audible. They remind us that collaboration with machines has always required attention, patience and restraint.

Perhaps the question is not whether these systems are computers, but whether we are willing to acknowledge computation when it takes place in steel rather than silicon.

We Stand Before the Machine: Rooms, Rules, Systems, and the Thinking Machines That Arrange Us

TIMOTHY ELLIM (TIMOTHY LIM)

When you enter a room, how do you know how to act?

Now imagine an empty room. Six white walls. Dust. Stagnant air.

Is there anything even in it?

I think there is.

The room is empty, but already filled with building regulations, rental agreements, fire codes, gallery etiquette, insurance policies, the implied achievement of a plinth, the directive authority of a spotlight. A room is not just walls. A room is a system of behaviour.

You may enter, but you cannot do anything you want. Your agency is already limited.

So how do you know how to act when strangers are present?

Perhaps the more uncomfortable question is: how do you know how to be you?

Now add the machine into the mix.

Not as a robot. Not as a cute cat hologram. Not even as a disembodied voice. The thinking machine arrives designed, built and packaged as sensor, camera, microphone, light, prompt, generative model, threshold, inference. It does not need to introduce itself. It does not need to say, "I am here."

It only needs to observe and act.

A room that can observe with the capacity to act is very different from a room that merely contains.

The thinking machine cannot become a human body yet. But it can be installed into the conditions around our bodies. It can become the way the room detects movement, hesitation, proximity, attention, participation.

Because rooms are made of rules, and the installed machine can become part of the rule.

So how does the machine know how to behave when it IS the room?

We, as humans, can enter rooms. We can be held by them. We can even navigate in the space of the room. But the machine can be installed into the systems of the room and define what the room can do in real time. The machine, unlike us, can become the room itself.

Imagine the machine is given several jobs at once: keep objects safe, guide visitors, encourage interaction, detect confusion, support cooperation, prevent misuse.

But how does it know what any of those words mean?

Does it read hesitation as hesitation, or refusal?

Laughter as comfort, or nerves?

Stillness as disengagement, or someone trying not to be seen failing?

By the time a human starts to speak, the room may already have made its first assumption.

Or perhaps the room has not made the assumption alone.

The designer, engineer, sensor, model,

lighting state, threshold, default, and inherited habit of interaction design are all folded into the room. The room does not decide like a person. It decides because it has been built to notice some things and not others.

Perhaps you, just like me, will realise that this can be a very big problem.

In these rooms, the machine may work through light, sound, delay, rhythm, atmosphere. A pulse can become instruction. A pause can become pressure. A glowing object can become a shared assumption.

Games have always known this.

Game systems argue through what they allow, block, repeat, reward, or punish. In a GenAI-mediated thinking machine embedded room, that argument leaves the screen and enters the social body. The machine does not need to say “stand there” if it can make standing there feel obvious.

You might know this feeling.

The moment in an installation when you are not sure whether you are playing, performing, testing, or being tested.

The moment you glance at a stranger to check whether you are doing it right.

We do not enter rooms as minds first. We enter as bodies. We turn, pause, mimic, hesitate, reach, withdraw. Before we can explain what we think, the body has already responded with, can I?

I first understood this through Lego.

At six, my friends and I sat around a box of bricks. We were together, but building on our own.

Then someone said, “Let’s build a spaceship.”

The bricks became parts of a shared problem. Someone built wings. Someone made a cockpit. Someone insisted it needed a laser cannon, or a door that opened badly but still counted.

The object became a way to play, argue, articulate, and become together.

We did not need to agree completely.

We only needed something shared enough to work with.

And quickly, norms formed.

Who leads. Who follows. Who waits. Who is seen as confident. Who is discriminated against. Who is punished for not understanding fast enough.

Rules are never cold logic. They contain and elicit emotions. Rules frustrate, seduce, shame, reward, and delight us.

In a GenAI-mediated room, the rule may not be written anywhere obvious, but it is still felt.

At its best, this pressure can become generous. A shared threshold. A mutual problem. A reason for strangers to coordinate. For a moment, people become a group.

But what feels intuitive to one person may feel hostile to another.

What looks like elegant coordination from outside may feel like pressure from within.

So before asking what GenAI can generate, I want to ask what it wants to arrange.

And the room does not have to be physical.

The room can be virtual, digital, imagined. A room is any container of information, behaviour, and consequence. A room is a system of rules.

So I ask one last time.

Now that there is a machine in the room, how do you know how to act?

Because if you do not think about this, the thinking machine will; and by design, it will also think for you.

Further Reading

I have included the following for further reading, and I wonder, once you have read them all, would you see the world in the way I do?

Berger, P.L. and Luckmann, T. (1966) *The social construction of reality*. New York: Doubleday.
Bogost, I. (2007) *Persuasive games: The expressive power of videogames*. Cambridge, MA: MIT Press.

Dolan, J. (2005) *Utopia in performance: Finding hope at the theater*. Ann Arbor: University of Michigan Press.

Gelfand, M.J., Gavrillets, S. and Nunn, N. (2024) ‘Norm dynamics’, *Annual Review of Psychology*, 75, pp. 341-378.

Haraway, D. (1988) ‘Situated knowledges’, *Feminist Studies*, 14(3), pp. 575-599.

Merleau-Ponty, M. (2012) *Phenomenology of perception*. London: Routledge.

Sicart, M. (2023) *The beautiful rule*. Cambridge, MA: MIT Press.

Simondon, G. (1980) *On the mode of existence of technical objects*. London, Ontario: University of Western Ontario.

Cover Image:

Paul Mazzitelli

NSEW - Taitograph

2025

Chisel nib marker pen on heavyweight Bristol paper

21 × 29.7 cm



Supported using public funding by

**ARTS COUNCIL
ENGLAND**

